

**Please read the manual before installation*

Professional Wind & Solar Hybrid (Street Light) Controller Manual



User Manual

Ver.4.0

Thank you for purchasing our product(s). The manual is provided to people who need to install and operate the controller. Read this manual before any work with controller and keep it carefully. The contents of this manual will be periodically updated or revised if necessary. However discrepancies cannot be excluded. Please refer to the actual product(s).

Symbols

The following symbols are used throughout this manual to indicate potentially dangerous conditions or mark important safety instructions.

 **WARNING:** Indicates a potentially dangerous condition. Use extreme caution when performing this task.

 **INDICATION:** Indicates a procedure or function that is important.

 **NOTE:** Indicates a specific description for content.

General Safety Information

- Before receive the product, check it carefully. Make sure whether the product is damaged during transport. If it is damaged, contact the shipping company or our company immediately.
- All installation and electrical work must only be performed by professional personnel.
- Without any professional guidance, do not disassemble or attempt to repair the controller.
- Do not use the controller without batteries.
- Do not cut off the connection of controller and batteries when controller is working normally.
- Keep children away from controller.
- Do not allow water to enter the controller.
- Confirm that power connections are tightened to avoid excessive heating from a loose connection. Make sure cables are suitable for system.

1 Product Introduction

This kind of wind and solar hybrid streetlight controller is special design for small off-grid wind solar hybrid generation system. Appearance is elegant, operations are easy. It also makes the course of which wind generator and solar panels charge to batteries safely and efficiently.

1.1 Functions and Features

1.1.1 Basic Functions

- **Wind Turbine and Load Adaptive Impedance Matching, maximize energy utilization.**

 There is internal resistance in Wind generators, batteries and loads. According to impedance matching principle, only when input impedance equals to output impedance, power utilization is maximal, get the maximum power.

- **Protect wind generator from over-revolution speed, over-voltage and over-current**

 Max revolution speed, max voltage and max current of wind generator could be set. Once the actual revolution speed, voltage or current over the set ones, PWM intelligent unloading will start automatically. That protect wind generator.

- **Intelligent limiting of batteries max current**

 Batteries maximum capacity could be set through this controller. According to the set maximum capacity, controller could calculate the maximum charging current. Then batteries will be protected.

- **Two DC output, each one has 9 output modes**

- **Function of manual brake**

- **Wind charging manual switch**

 On the controller you can manually set whether using wind charge to battery.

- **Solar charging manual switch**

 On the controller you can manually set whether using solar charge to battery.

- **DC output manual switch**

 Users could manually set whether use DC output.

- **RS232 /RS485 interface**

 By serial interface communication, you could monitor the whole system, storage and analyze data.

Program could be upgraded by serial interface.

Connect PC and controller by serial interface. You could set the parameters on PC and controller simultaneously.

 Software is free, easy to operate and no need to be installed.

➤ **BOOST and BUCK function in one**

📖 (If do not buy Boost & Buck Wind Solar Streetlight Controller, there is no this function)
Once wind generator voltage is lower than battery voltage, controller starts boost module automatically. Wind generator voltage is increased to the charging voltage, and it is boost charging. When wind generator voltage is higher than battery voltage, in order to acquire max power, buck module of controller will be started, the generator is buck charging.

Loads lower the revolution speed of wind generator, when it is breeze. That decreases the output power of wind generator. Through max current tracking (MCT) and max power point tracking (MPPT), output of wind generator is stabilized at the max balance of wind energy utilization. Combine with boost and buck function, wind energy utilization is increased.

➤ **BOOST function**

(If do not buy Boost Wind Solar Streetlight Controller, there is no this function)

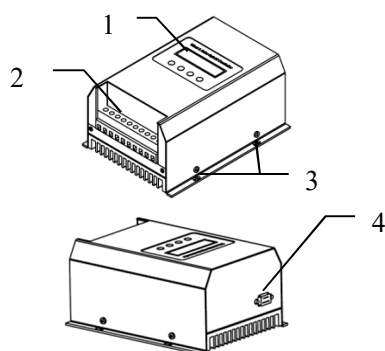
📖 Once wind generator voltage is lower than battery voltage, controller starts boost module automatically to charge to battery.

➤ **BUCK function**

(If do not buy Buck Wind Solar Streetlight Controller, there is no this function)

📖 Once wind generator voltage is higher than battery voltage, controller starts buck module automatically to charge to battery.

1.2 Appearance



Item	Name	Description
1	LCD display panel	A friendly human-computer interaction interface. Running data and configuration parameters are displayed in the LCD screen. Parameters could be set by keys on the panel.
2	Terminal block	Connect wind generator, pv panel, battery and load.
3	Mounting holes	Install controller
4	RS232 interface	Communication interface (If don't purchase, no this interface)

Figure1-1 Appearance description

1.3 Dimensions

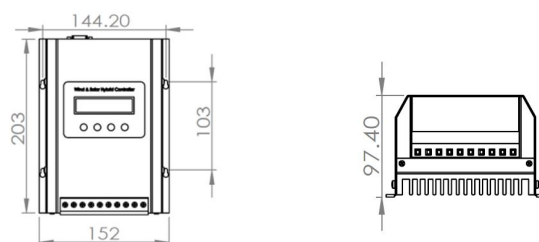


Figure1-2 Dimensions of controller

2 Installation and Electrical Connection

2.1 Installation

2.1.1 Mounting Notes

- ⚠ Read through this entire section first before beginning installation.
- ⚠ All mounting work must only be performed by professional personnel.
- ⚠ Disconnect all sources of power to the controller before installing or adjusting.
- ⚠ Do not allow water and snows enter the controller.
- ⚠ Install in locations where is dustless, airy and avoid direct sunlight.
- ⚠ If install controller in a cabinet, make sure there is enough space for controller heat-dissipating.
- ⚠ Keep controller away from corrosive gas and intense electromagnetic interference.
- ⚠ Locate the product in where easy to install, electrical connection and service.

2.1.2 Mounting Steps

1. Choose mounting location.(Please refer to installation notes)
2. Check for clearance around the location; make sure there is enough space for connecting cables.
3. Prepare tools for installation.
4. Place the controller to the mounting location.
5. Check that the controller is securely mounted.

2.2 Electrical Connection

2.2.1 Overview of Electrical Connection

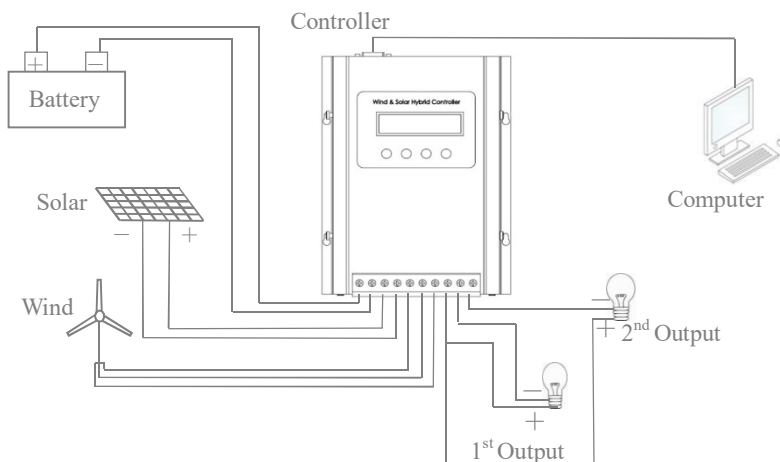


Figure2-1 Overview of electrical connection

2.2.2 Wiring Notes

- ⚠ Improper operation during the wiring process can cause fatal injury to the operator or unrecoverable controller damage.
- ⚠ Only qualified personnel can perform the wiring work.
- ⚠ All cables must be undamaged, properly insulated and adequately dimensioned.
- ⚠ Make sure that all cables are firmly attached. Unsecured cables create loose and resistive connections which may lead to excessive heating and /or fire.
- ⚠ For mobile applications, be sure to secure all wiring, avoid loose connections.

2.2.3 Wiring Steps (Follow the bellow suggestions and steps to connect)

2.2.3.1 Battery Wiring

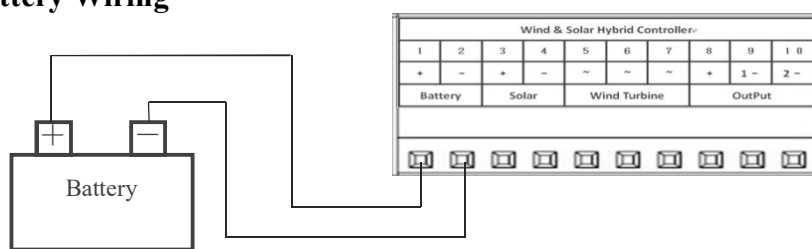


Figure2-2 Battery wiring

Connect battery positive(+) and negative(-) wires to controller as shown in figure2-2.



Be careful of avoiding short circuit when wiring the battery.



Although controller has the protection of battery anti-reverse, but anti-connecting of positive (+) and negative(-) is forbidden.

2.2.3.2 Solar Wiring

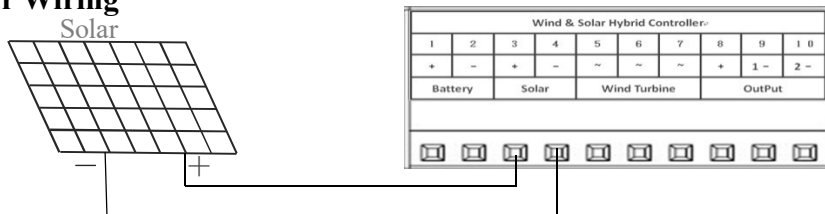


Figure2-3 Solar wiring

- ⚠ Connect solar positive(+) and negative(-) wires to controller as shown in figure2-3.
- ⚠ The solar PV array may produce high voltages in sunlight. Be careful of electric shock when wiring.
- 🔒 Although controller has the protection of solar anti-reverse, but anti-connecting of positive (+) and negative(-) is forbidden.

2.2.3.3 Wind Generator Wiring

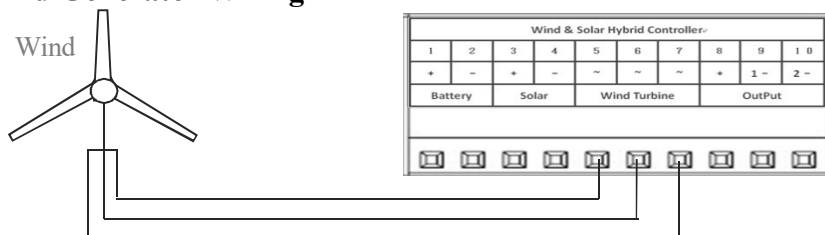


Figure2-4 Wind generator wiring

- ⚠ Connect wind generator wires to controller as shown in figure2-4.
- ⚠ The wind generator could produce high voltages. Be careful of electric shock.
- ⚠ When it is breeze or windless, connection of wind generator and controller would be safer and better. Only when controller is in the state of start-up, high-speed rotate wind generator could be connected.

2.2.3.4 Loads Wiring

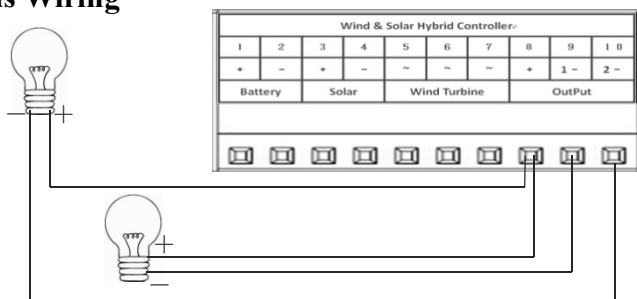


Figure2-5 Loads wiring

- ⚠ Connect loads positive(+) and negative(-) wires to controller as shown in figure2-5.
- ⚠ Be careful of avoiding short circuit when wiring loads.
- 🔒 This controller have two outputs, they are common anode. Each load must be direct current. Each load rated voltage and battery rated voltage should be the same.
- 🔒 Anti-connecting of loads positive (+) and negative(-) is forbidden.
- 🔒 Loads current should not exceed the rating loads current.

2.2.4 Confirm Wiring

Double-check the wiring. Make sure each connection is correct. Secure no loose and resistive connections.

3 Operation

3.1 Description of Buttons

Buttons	Description
Menu	Enter into sub-screen or confirm the command.
◀	Switch between sibling menu or decrease the setting value.(Press more than two seconds change the setting value quickly)
▶	Switch between sibling menu or increase the setting value.(Press more than two seconds change the setting value quickly)
Esc	Return to parent screen or cancel the command.

3.2 Overview of LCD Menu

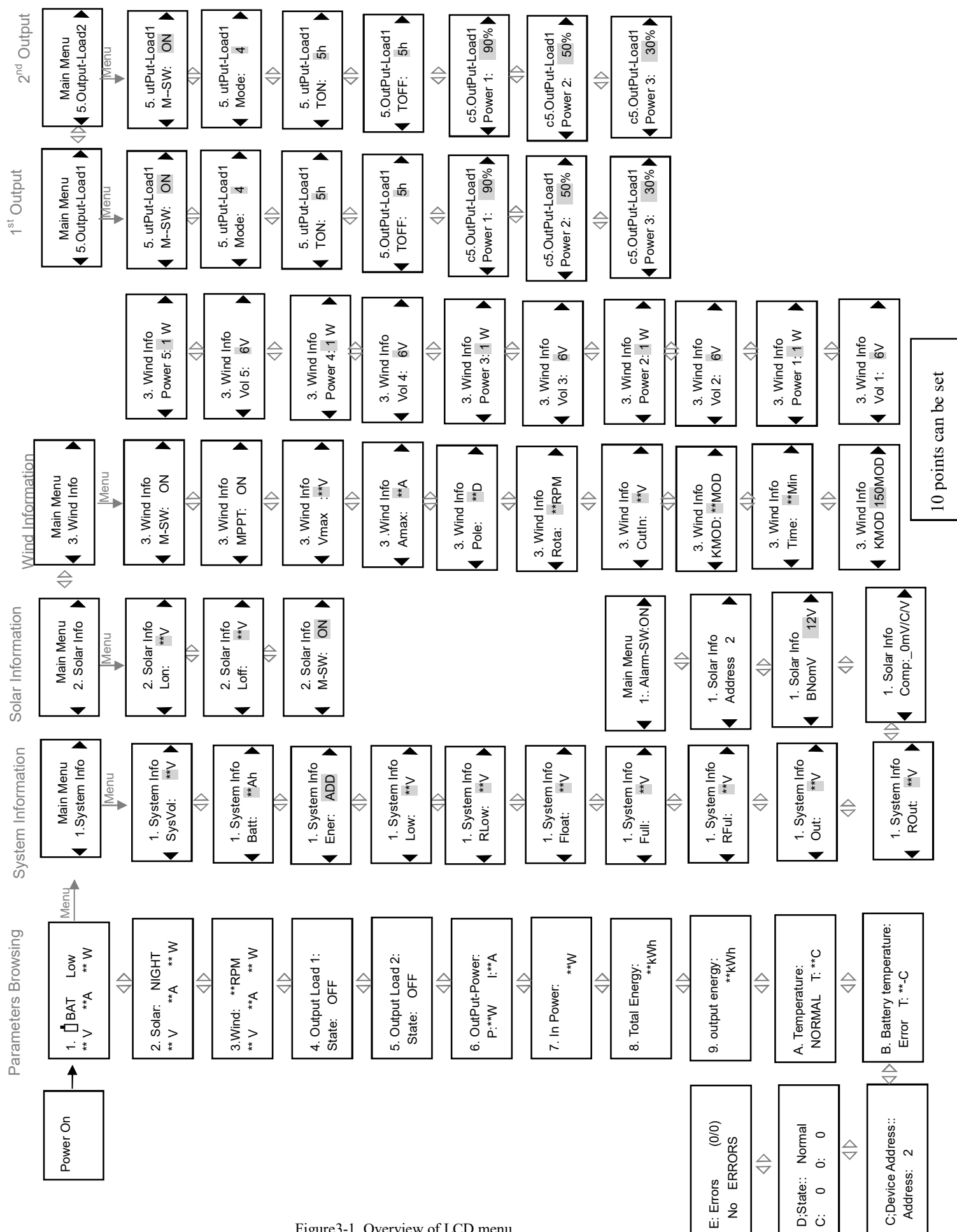


Figure3-1 Overview of LCD menu

 Some parameters could be set manually by users.

 Parameter setting steps:

1. Press “Menu” to enter the setting state, setting value would flash.
2. Press “◀” to decrease the setting value. Press “▶” to increase the setting value.
3. The number would be set one by one from left to right. After one number is set, press “Menu”, the flash number will be the next. When the last number (unit) is finished press “Menu” to save the setting, press “Esc” to back out.

3.3 Parameters Browsing

3.3.1 Battery

1.  : BAT Low
** V ** A ** W

 **%—Percentage of battery power

 Right corner have below displays:

 **Low**—Battery power is low

Normal—Battery power is normal

Full—Battery power is full

 **V—Battery voltage

 **A—Battery charging current

 **W—Battery charging power

3.3.2 Solar (if not purchase solar charge ,please disregard it)

2. Solar : NIGHT
** V ** A ** W

 Right corner have below displays:

DAY— When solar voltage is higher than “Lon” voltage, right corner display “DAY”. It delays 1 minute. When controller is power on, it displays “DAY”, analysis present circumstance, 1 minute later it displays normally.

NIGHT—When solar voltage is lower than “Loff” voltage, right corner display “NIGHT”. It delays 1 minute.

 **V—Solar charging voltage

 **A—Solar charging current

 **W—Solar charging power

3.3.3 Wind

3. Wind: **RPM
** V ** A ** W

 **RPM— rotating speed of wind generator. (Normal working state is this display)

 Other displays:

DumpWork—when battery is about full, and the wind turbine does not dump load completely;

Brake-ON—“Brake” in wind information is “ON”

MS-OFF—“M-SW” in wind information is “OFF”

Rmax***—when rotate speed of wind generator is greater than “Rota”, generator is braking. Braking “Time” is countdown.

“Rota”, “Time” could be set in 3.6.

 **V—Wind input voltage

 **A—Wind input current

 **W—Wind input power

3.3.4 Output

4. Output Load 1:
State: OFF



Output Load 1—1st output.

State: OFF

Output is off.

State: HALF

Output is working on half power

State: FULL

Output is working on full power

5. Output Load 2:
State: OFF



Output Load 1—1st output.

State: OFF

Output is off.

State: HALF

Output is working on half power

State: FULL

Output is working on full power

6. Output Power:
P: **W I: **A



P—output power



I—output current

3.3.5 Input

7. In Power:
**W



In-Power—total input power

3.3.6 Total Generated Energy

8. Total Energy
**kWh



Total-Energy—total generated energy

This value is cumulative. If you want to start from 0, set "Ener" in 3.4

3.3.7

9. Output Energy
**kWh



Out put energy—total power

This value is cumulative. If you want to start from 0, set "Ener" in 3.4. Total energy will be clear at the same time

3.3.8 Temperature Protection

A. Temperature:
NORMAL T: **C



TEMPERATURE—Temperature protection.



NORMAL: MOS fet temperature is normal.



HIGH: MOS fet temperature is high.

OTP: MOS fet temperature is over the protect temperature.

ERROR: Temperature detection is error.

****C:** ** Celsius degree.

B.BATT Temperature:
Error T: **C

TEMPERATURE—Temperature protection.

NORMAL: Battery temperature is normal.

HIGH: Battery temperature is high.

OTP: Battery temperature is over the protect temperature.

ERROR: Temperature detection is error.

****C:** ** Celsius degree.

3.3.9 Device Address

C. Device Address
Address: 2

3.3.10 State: Normal

D. State: Normal
C: 0 0: 0

3.3.7 Error Code

A. Errors: (0/0)
No Errors.

0/0 —For example: If it is 2/4. 2 indicate the error display on next line is the second error of the 4 errors. 4 indicate there are 4 errors in all now.

Press “Menu” key to browsing what error code display on next line, press and to switch over.

When the second line display “No Errors”, press “Menu” key to into parameters browsing.

NO BATTERY—battery is not detected or battery voltage is too low

SHORT LOAD—load short-circuit or load current is large

SOLAR OVER V—solar voltage is too high

LOAD OVER V—load voltage is too high

TEMP ERROR—temperature sensor error

EEPROM ERROR.—data storage error

No Errors—normal

3.4 System Information

Main Menu
◀ 1. System Info ▶

System Info—system information

1. System Info
◀ SysVol: **V ▶

SysVol—Rated voltage of battery bank

1. System Info
◀ Batt: **Ah ▶

Batt—battery capacity

1. System Info
◀ Energy: ADD ▶

Energy—ADD: Energy accumulates
CLEAR: Energy clears to 0

1. System Info
◀ Low: 10.2v ▶

Low—under voltage

1. System Info
◀ RLow: **V ▶

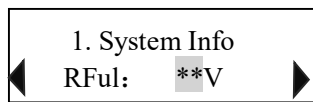
RLow—under voltage recovery voltage

1. System Info
◀ Float: **V ▶

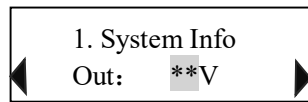
Flot—floating voltage

1. System Info
◀ Full: **V ▶

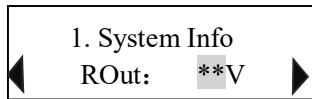
Full—over voltage



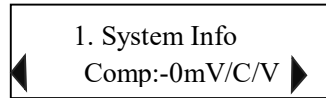
RFul—over voltage recovery voltage



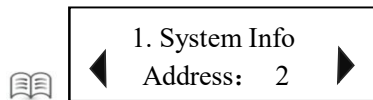
Out—over load protection voltage



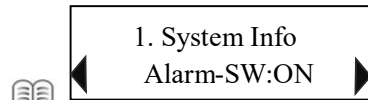
ROut—over-load recovery voltage



Will adjust the battery voltage automatically according to the environment temperature

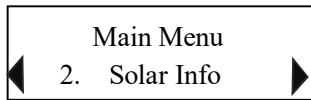


RS485 address

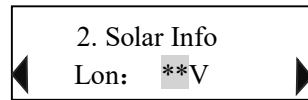


When controller working abnormal .It will alarm

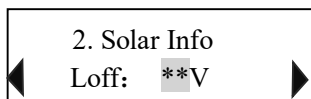
3.3.8 Solar Information (if not purchase solar charge ,please disregard it)



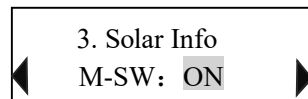
Solar Info—solar information



Lon—light control on voltage

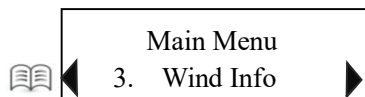


Loff—light control off voltage

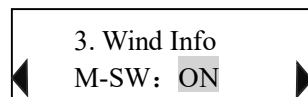


M-SW—wind charging manual switch
ON: turn on OFF: turn off

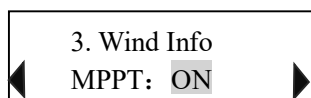
3.5 Wind Information



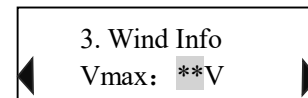
Wind Info—wind information



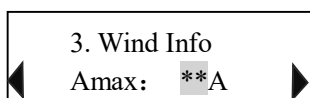
M-SW—wind charging manual switch
ON: turn on OFF: turn off



MPPT—wind MPPT function switch.
ON: turn on OFF: turn off
If do not purchase the function of Boost/Buck/Boost & Buck, it would display “NONE”.



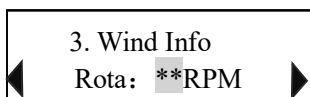
Vmax—Maximum voltage of wind generator



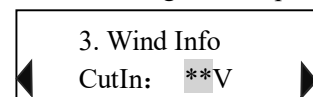
Amax—Maximum current of wind generator



Pole—wind generator pole logarithm



Rota—maximum rotate speed of wind generator.



CutIn—wind start charging voltage
If do not purchase the function of Boost or Boost & Buck, it would display “NONE”

3. Wind Info

Time: **min

Time—braking time

3. Wind Info

Vol 1: **V

3. Wind Info

KMOD: **MOD

KMOD—MPPT correction factor
(Please do not modify this parameter)

3. Wind Info

Power 1: **W

There are 10 points can be set ,User's can set the points based on the power curve of the wind turbine

3.6 1st Output Information

Main Menu

4. OutPut-Load1

OutPut-Load1—1st output

4. OutPut-Load1

Mode: ** 1/4

Mode—1st output mode

4. OutPut-Load1

Toff: **H 3/4

Toff—1st output time control off

4. OutPut-Load1

Power2: 100%

Percentage of the power for the output load on second time period

4. OutPut-Load1

M-SW: ON 4/4

M-SW—1st output manual switch
ON: turn on 1st output; OFF: turn off 1st output

4. OutPut-Load1

Ton: **H 2/4

Ton—1st output time control on

4. OutPut-Load1

Power1: 100%

Percentage of the power for the output load on first time period

4. OutPut-Load1

Power3: 100%

Percentage of the power for the output load on third time period

3.7 2nd Output Information

Main Menu

4. OutPut-Load1

OutPut-Load1—1st output

4. OutPut-Load1

Mode: ** 1/4

Mode—1st output mode

4. OutPut-Load1

Toff: **H 3/4

Toff—1st output time control off

4. OutPut-Load1

Power2: 100%

Percentage of the power for the output load on second time period

4. OutPut-Load1

M-SW: ON 4/4

4. OutPut-Load1

Ton: **H 2/4

Ton—1st output time control on

4. OutPut-Load1

Power1: 100%

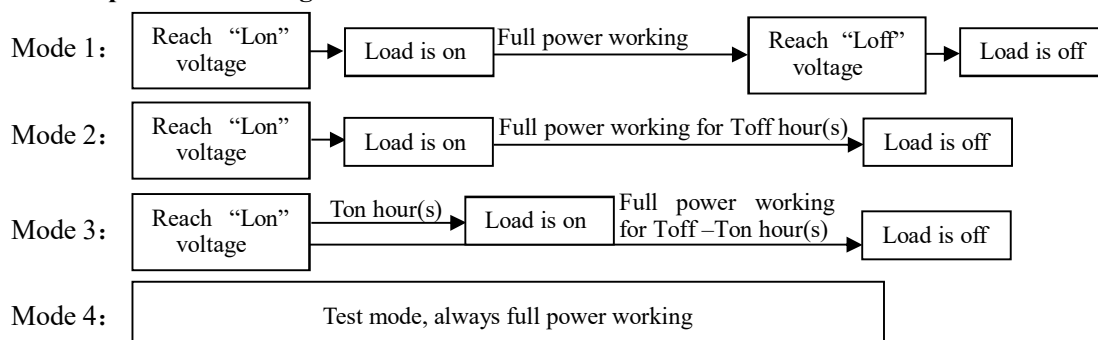
Percentage of the power for the output load on first time period

4. OutPut-Load1

Power3: 100%

Percentage of the power for the output load on third time period

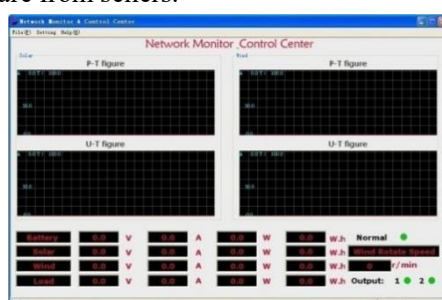
Each output has following 9 modes for users to set:



4 Software

The software is easy to operate need not to be installed. You can browse and set parameters on PC through the software. Users could ask the software from sellers.

- Browsing interface on PC:



- Contents displayed on browsing interface:
 Battery: voltage; charging current; power; power obtained; generated energy obtained.
 Solar: voltage; charging current; charging power; generated energy.
 Wind turbine: voltage; charging current; charging power; generated energy.
 Output load: voltage; current; power; output energy.
- Software using method could reference to the instruction of software compressed file.

5 Warranty

The product is warranted for one year from the date of shipment to the original end user. During warranty period, if failure occurs when the product normal using, our company will repair or replace the failure product.

Out of warranty period, we supply repair service, but for charges.

This warranty is only provided to buyers who have bought the product and signed the CI with us, and the warranty is nontransferable.

Our company reserves the right to change products and without notice when products update.

This warranty does not apply under the following conditions:

- Damage by not operating in accordance with user manual.
- Damage by accident, negligence, abuse or improper use.
- Unauthorized product modification or attempted repair.
- Damage occurring during shipment

Parameters

Wind Solar Hybrid Streetlight Controller (Boost & Buck)

Model	TGWS200W-12V	TGWS300W-12V	TGWS400W-12V	TGWS500W-12V
Rated wind power	100W	200W	300W	300W
Rated solar power	100W	100W	100W	200W
Nominal system voltage	12V			
Max. solar input voltage	40V			
Under voltage (Low)	10.2V(adjustable)			
Under voltage recovery voltage(Rlow)	11.5V(adjustable)			
Over voltage(Full)	14.7V(adjustable)			
Over voltage recovery voltage(RFull)	23.2V(adjustable)			
Float voltage(Flot)	23.5V(adjustable)			
Over-load voltage(Out)	17.7V(adjustable)			
Over-load recovery voltage(ROut)	15.0V(adjustable)			
Wind dumpload rotate speed(Rota)	500R(adjustable)			
Wind pole logarithm(Pole)	10D(adjustable)			
Wind charging range	DC (10-40)V			
Wind start charging voltage(CutIn)	8V(adjustable)			
Wind dumpload voltage(Vmax)	40V(adjustable)			
Solar voltage of light-control on (Lon)	6V(adjustable)			
Solar voltage of light-control off (Loff)	6V(adjustable)			
Rated output current	15A			
1st output working mode(Mode)	Mode 1: Light-control on. Light-control off (adjustable)			
2nd output working mode(Mode)	Mode 2: Light-control on. Full power working for 5 hours, light off(adjustable)			
Dump load control mode	Over rotate speed limiting, Over voltage limiting, Over Current limiting, PWM			
Wind charging mode	MPPT(Boost & Buck) & PWM			
Solar charging mode	PWM (no purchase ,no this function)			
Display mode	LCD			
Display content	Battery: voltage;charging current; Percentage of battery power. Wind: voltage; charging current; rotate speed; output current; output power Solar: voltage; charging current. Loads: current; power; working mode. System: state; generated energy; error code			
Operating temperature & Relative humidity	- 20~ + 55°C/35~85%RH(Non-condensing)			
Quiescent power drain	≤3W			
Protection type	Battery: over-discharge protection; over-charge protection; anti-reverse connection. Wind: Over rotate speed protection, over voltage protection, over current protection. Loads: Over-load protection			
Controller size	203mm*152mm*97.4mm			
Package size	290mm*170mm*190mm			
Net weight	2.1Kg			
Gross weight	2.5Kg			
Communication function	RS485			

Model	TGWS200W-24V	TGWS300W-24V	TGWS400W-24V	TGWS500W-24V	TGWS800W-24V
Rated wind power	100W	200W	300W	300W	500W
Rated solar power	100W	100W	100W	200W	300W
Nominal system voltage	24V				
Max. solar input voltage	160V				
Under voltage (Low)	20.4V(adjustable)				
Under voltage recovery voltage(Rlow)	23.0V(adjustable)				
Over voltage(Full)	29.4V(adjustable)				
Over voltage recovery voltage(RFull)	26.4V(adjustable)				
Float voltage(Flot)	27.0V(adjustable)				
Over-load voltage(Out)	35.5V(adjustable)				
Over-load recovery voltage(ROut)	30.0V(adjustable)				
Wind dumpload rotate speed(Rota)	500R(adjustable)				
Wind pole logarithm(Pole)	10D(adjustable)				
Wind charging range	DC (10-80)V				
Wind start charging voltage(CutIn)	10V(adjustable)				
Wind dumpload voltage(Vmax)	80V(adjustable)				
Solar voltage of light-control on (Lon)	6V(adjustable)				
Solar voltage of light-control off (Loff)	6V(adjustable)				
Rated output current	15A				
1st output working mode(Mode)	Mode 1: Light-control on. Light-control off (adjustable)				
2nd output working mode(Mode)	Mode 2: Light-control on. Full power working for 5 hours, light off(adjustable)				
Dump load control mode	Over rotate speed limiting, Over voltage limiting, Over Current limiting, PWM				
Wind charging mode	MPPT(Boost & Buck) & PWM				
Solar charging mode	PWM (no purchase ,no this function)				
Display mode	LCD				
Display content	Battery: voltage;charging current; Percentage of battery power. Wind: voltage; charging current; rotate speed; output current; output power Solar: voltage; charging current. Loads: current; power; working mode. System: state; generated energy; error code				
Operating temperature & Relative humidity	- 20~ + 55°C/35~85%RH(Non-condensing)				
Quiescent power drain	≤3W				
Protection type	Battery: over-discharge protection; over-charge protection; anti-reverse connection. Wind: Over rotate speed protection, over voltage protection, over current protection. Loads: Over-load protection				
Controller size	203mm*152mm*97.4mm				
Package size	290mm*170mm*190mm				
Net weight	2.1Kg				
Gross weight	2.5Kg				
Communication function	RS485				

Model	TGWS400W-48V	TGWS500W-48V	TGWS800W-48V	TGWS1000W-48V	TGWS1200W-48V
Rated wind power	300W	300W	500W	800W	1000W
Rated solar power	100W	200W	300W	200W	200W
Nominal system voltage	48V				
Max. solar input voltage	160V				
Under voltage (Low)	40.8V(adjustable)				
Under voltage recovery voltage(Rlow)	46.0V(adjustable)				
Over voltage(Full)	58.8V(adjustable)				
Over voltage recovery voltage(RFull)	52.8V(adjustable)				
Float voltage(Flot)	54.0V(adjustable)				
Over-load voltage(Out)	65V(adjustable)				
Over-load recovery voltage(ROut)	60.0V(adjustable)				
Wind dumpload rotate speed(Rota)	500R(adjustable)				
Wind pole logarithm(Pole)	10D(adjustable)				
Wind charging range	DC (10-80)V				
Wind start charging voltage(CutIn)	10V(adjustable)				
Wind dumpload voltage(Vmax)	80V(adjustable)				
Solar voltage of light-control on (Lon)	6V(adjustable)				
Solar voltage of light-control off (Loff)	6V(adjustable)				
Rated output current	15A				
1st output working mode(Mode)	Mode 1: Light-control on. Light-control off (adjustable)				
2nd output working mode(Mode)	Mode 2: Light-control on. Full power working for 5 hours, light off(adjustable)				
Dump load control mode	Over rotate speed limiting, Over voltage limiting, Over Current limiting, PWM				
Wind charging mode	MPPT(Boost & Buck) & PWM				
Solar charging mode	PWM (no purchase ,no this function)				
Display mode	LCD				
Display content	Battery: voltage;charging current; Percentage of battery power. Wind: voltage; charging current; rotate speed; output current; output power Solar: voltage; charging current. Loads: current; power; working mode. System: state; generated energy; error code				
Operating temperature & Relative humidity	- 20~ + 55°C/35~85%RH(Non-condensing)				
Quiescent power drain	≤3W				
Protection type	Battery: over-discharge protection; over-charge protection; anti-reverse connection. Wind: Over rotate speed protection, over voltage protection, over current protection. Loads: Over-load protection				
Controller size	203mm*152mm*97.4mm				
Package size	290mm*170mm*190mm				
Net weight	2.1Kg				
Gross weight	2.5Kg				
Communication function	RS485				

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